

Test Report Data Processing on Modern Hardware Assignment2

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Evaluate volano model

- we comment out operator at a time and only test vulano performance

Samples: 21K of event 'cycles:Pu', Event count (approx.): 1993383978

Overhead	Command	Shared Object	Symbol
82.75%	weedb	weedb	[.] SelectionOp::next()
10.74%	weedb	weedb	[.] AggregationOp::next()
6.38%	weedb	weedb	[.] ScanOp::next()
0.11%	weedb	[unknown]	[k] 0xfffffffffa2000be0
0.01%	weedb	ld-linux-x86-64.so.2	[.] do_lookup_x
0.00%	weedb	libc.so.6	[.] __strchr_avx2
0.00%	weedb	ld-linux-x86-64.so.2	[.] strcmp
0.00%	weedb	[unknown]	[k] 0xfffffffffa2000080
0.00%	weedb	ld-linux-x86-64.so.2	[.] _dl_start

- High IPC, High Branch-Hit, 1.000 CPUs utilized

```

5296.18 msec task-clock:u          # 1.000 CPUs utilized
      0 context-switches:u        # 0.000 /sec
      0 cpu-migrations:u          # 0.000 /sec
    24543 page-faults:u           # 4.634 K/sec
19838084427 cycles:u              # 3.746 GHz
40811445513 instructions:u       # 2.06 insn per cycle
7220275647 branches:u           # 1.363 G/sec
11783074 branch-misses:u         # 0.16% of all branches

```

5.296865778 seconds time elapsed

5.238871000 seconds user

0.049989000 seconds sys

- Most time spend on `::next()`, very costly switching between function calls
- Maximum resident set size (kbytes): 1566268
- 73.49% cache misses

```

41338529 cache-misses:u          # 73.49% of all cache ref
s
56248475 cache-references:u
      0 major-faults:u

```

5.136075542 seconds time elapsed

5.058617000 seconds user

0.069980000 seconds sys

- call graph

Samples: 23K of event 'cycles:Pu', Event count (approx.): 20330226509

	Children	Self	Command	Shared Object	Symbol
+	98.65%	11.05%	weedb	weedb	[.] AggregationOp::next()
-	98.21%	0.00%	weedb	weedb	[.] _start
					_start
					__libc_start_main_impl (inlined)
					__libc_start_call_main
					main
					execVolcano(RelOperator*)
					PullDriver::volcano(RelOperator*, Relation*)
					- AggregationOp::next()
					- SelectionOp::next()
					- 71.83% SelectionOp::next()
					- 56.67% SelectionOp::next()
					- 42.30% SelectionOp::next()
					- 26.33% SelectionOp::next()
					6.24% ScanOp::next()

Evaluate materialization model

- High IPC. High Branch hit 0.780 CPUs utilized

```

5043.57 msec task-clock:u          # 0.780 CPUs utilized
    0   context-switches:u        # 0.000 /sec
    0   cpu-migrations:u          # 0.000 /sec
 38001   page-faults:u            # 7.535 K/sec
11245638880   cycles:u            # 2.230 GHz
24720355408   instructions:u      # 2.20 insn per cycle
1151603052    branches:u          # 228.331 M/sec
 10321367     branch-misses:u     # 0.90% of all branches

6.462962754 seconds time elapsed

3.651745000 seconds user
1.428056000 seconds sys

```

- faster than volcano model
- Maximum resident set size (kbytes): 3128164
- Almost no overhead

Samples: 16K of event 'cycles:Pu', Event count (approx.): 12337950958

Overhead	Command	Shared Object	Symbol
75.95%	weedb	weedb	[.] compareNotEquals(long*, long, long*, unsigned long)
14.48%	weedb	weedb	[.] scanLong(long*, long*, unsigned long)
9.34%	weedb	weedb	[.] aggSum(long*, unsigned long)
0.20%	weedb	[unknown]	[k] 0xfffffffffa200be0
0.01%	weedb	ld-linux-x86-64.so.2	[.] _dl_relocate_object
0.01%	weedb	ld-linux-x86-64.so.2	[.] do_lookup_x
0.01%	weedb	[unknown]	[k] 0xfffffffffa200f30
0.00%	weedb	ld-linux-x86-64.so.2	[.] __tunable_get_val
0.00%	weedb	ld-linux-x86-64.so.2	[.] _dl_cache_libcmp
0.00%	weedb	ld-linux-x86-64.so.2	[.] _GI__tunables_init
0.00%	weedb	ld-linux-x86-64.so.2	[.] _start

- Cache-Miss 62.97%

```

445873453  cache-misses:u          # 62.97% of all cache re
fs
708125315  cache-references:u
0    major-faults:u

4.839801004 seconds time elapsed

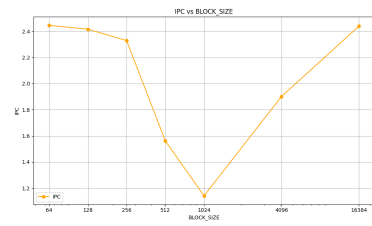
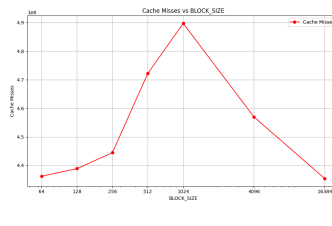
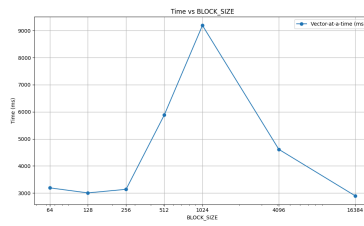
3.907042000 seconds user
0.929296000 seconds sys

```

Observations

- Materialization model cuts the execution instructions by half, but double the memory cost
- Materialization model reduces cache-miss by batching
- In general, materialization model more faster
- One main conclusion is that: tuple-at-a-time increase cost linearly, operator-at-a-time amortized the cost

Evaluate the influence of block-size on vector-at-a-time model



- We observe that the worst performance occurs in Block size 1024, after which it gains better
- the cache efficiency is improving but it will never exceed the performance of operator-at-a-time, because operator-at-a-time has higher throughput